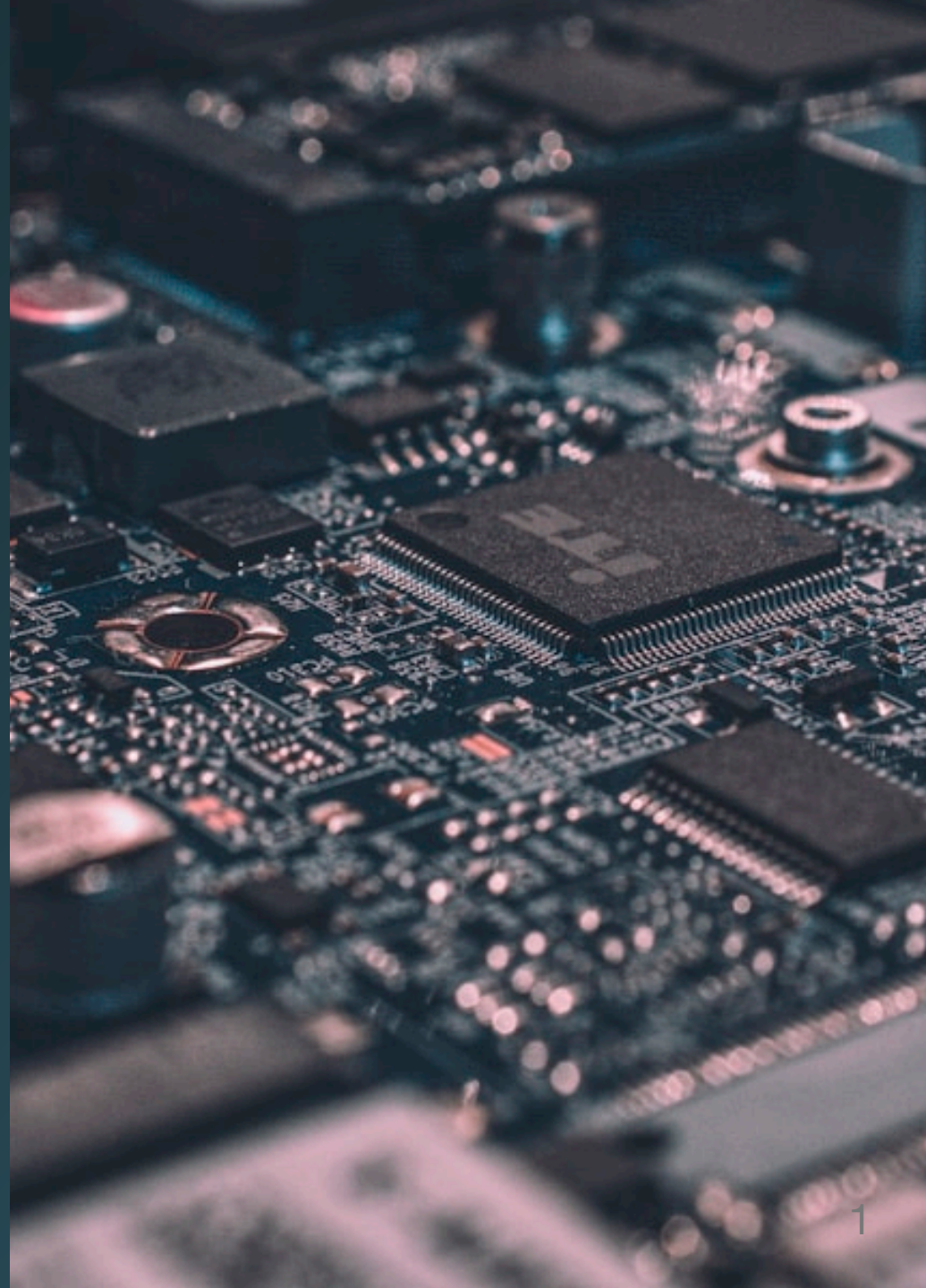
Scrape Pipeline

Lesson 09 · ISBA 4715



Agenda

	Part	What
1	**Scraping Concepts**	What it is, etiquette, the tools we'll use *(slides)*
2	**Python Pipeline**	Tavily + Firecrawl, saving to `knowledge/raw/` *(live code)*
3	**MCP Upgrade**	Claude Code as the pipeline *(demo)*
4	**Your Project**	Scrape at least one source into your repo

How did Google build its first index?

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Web scraping — reading pages programmatically.

Web Scraping

> Gathering data from websites **without an API** and **without a human clicking around in a browser**

Why Scrape?

No API exists for this data

Constantly-changing list (press releases, bios, news)

The API is paywalled

You want to automate something a human would do

Etiquette and Caveats

Rule	Why
Check `robots.txt`	Sites declare what bots can crawl
Rate limit — `time.sleep()`	Don't hammer the server
Respect Terms of Service	Legal + ethical obligation
Expect layout changes	HTML is not a stable contract
Be mindful of PII	Don't scrape what people did not consent to share

Where does scraped data go in THIS project?

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MP03 (API)

request → CSV → SQL warehouse

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→

MP04 (Scrape)

extract → markdown → knowledge/raw/

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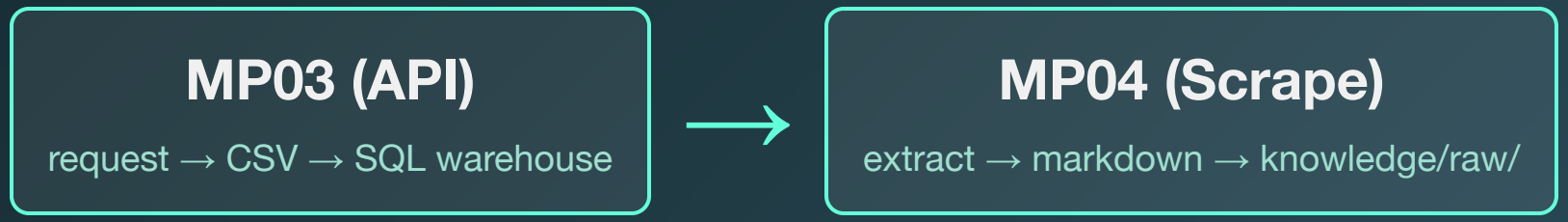


These feed your **knowledge base** — the unstructured side of your portfolio project.

The Two-Step Pattern

Discover: find URLs worth scraping → **Tavily**

Extract: turn each URL into clean markdown → **Firecrawl**

Tavily — AI-Native Search API

You send: a **query**

You get back: **JSON** with ranked URLs, titles, content previews

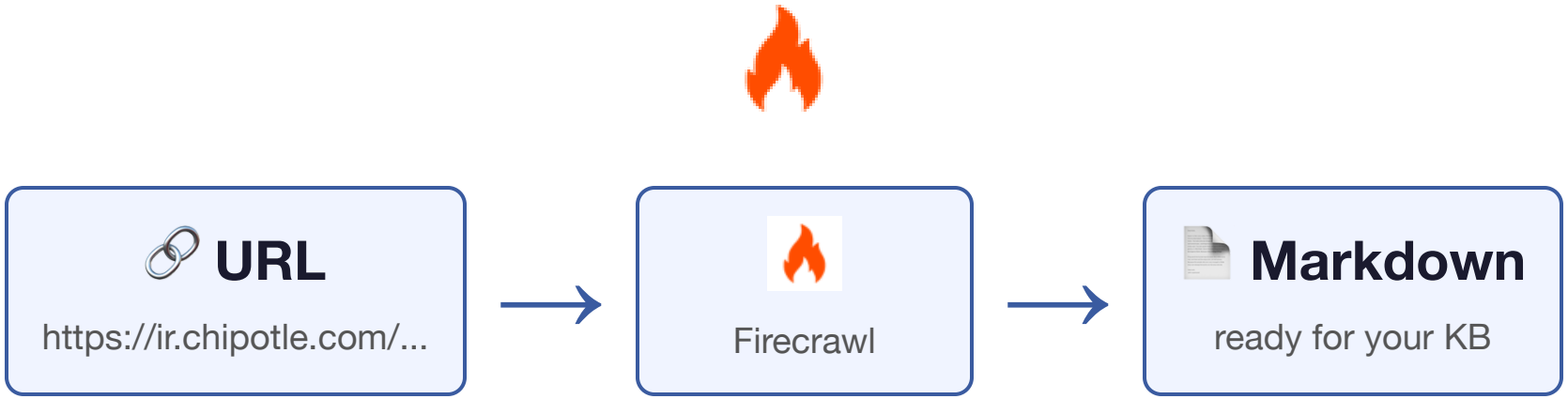
```
{"results": [
  {"url": "...", "title": "...", "content": "...", "score": 0.87},
  ...
]}
```

Tavily vs. Claude Code's WebSearch

Tavily (API)	WebSearch (chat)
Returns JSON	Returns a paragraph
For your pipeline	For you
Runs in GitHub Actions	Requires interactive Claude Code
Reproducible	Each answer is different

> If a cron job needs the answer, use the API. If a human needs the answer, use the chat tool.

Firecrawl — URL → Clean Markdown



No BeautifulSoup. No `select` or `find`. No DOM inspection.

Last Year We Taught BeautifulSoup

- Parse HTML tag-by-tag
- Hunt for the right CSS selector
- Break every time the site redesigns

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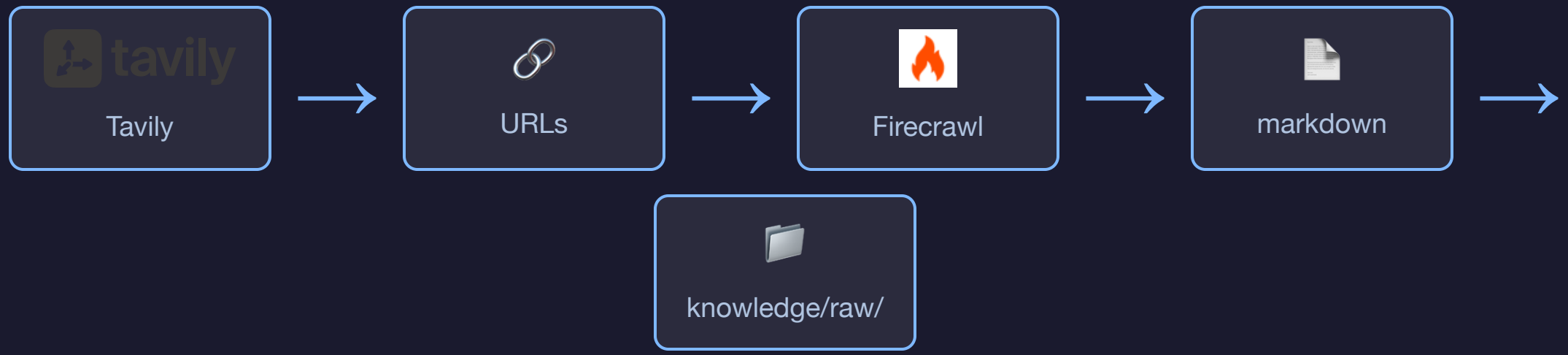
This year we don't. Firecrawl does all three in one call, and handles JavaScript rendering too. If you want BeautifulSoup for interview prep, read the docs — you do not need it for this project.

Build vs. Buy — Scraping Edition

Problem	Don't	Do
🔍 Find relevant URLs	Write a crawler	**Tavily**
📄 Turn URL into text	Write a parser	**Firecrawl**
🖱️ Click through a login	Write a bot	Playwright

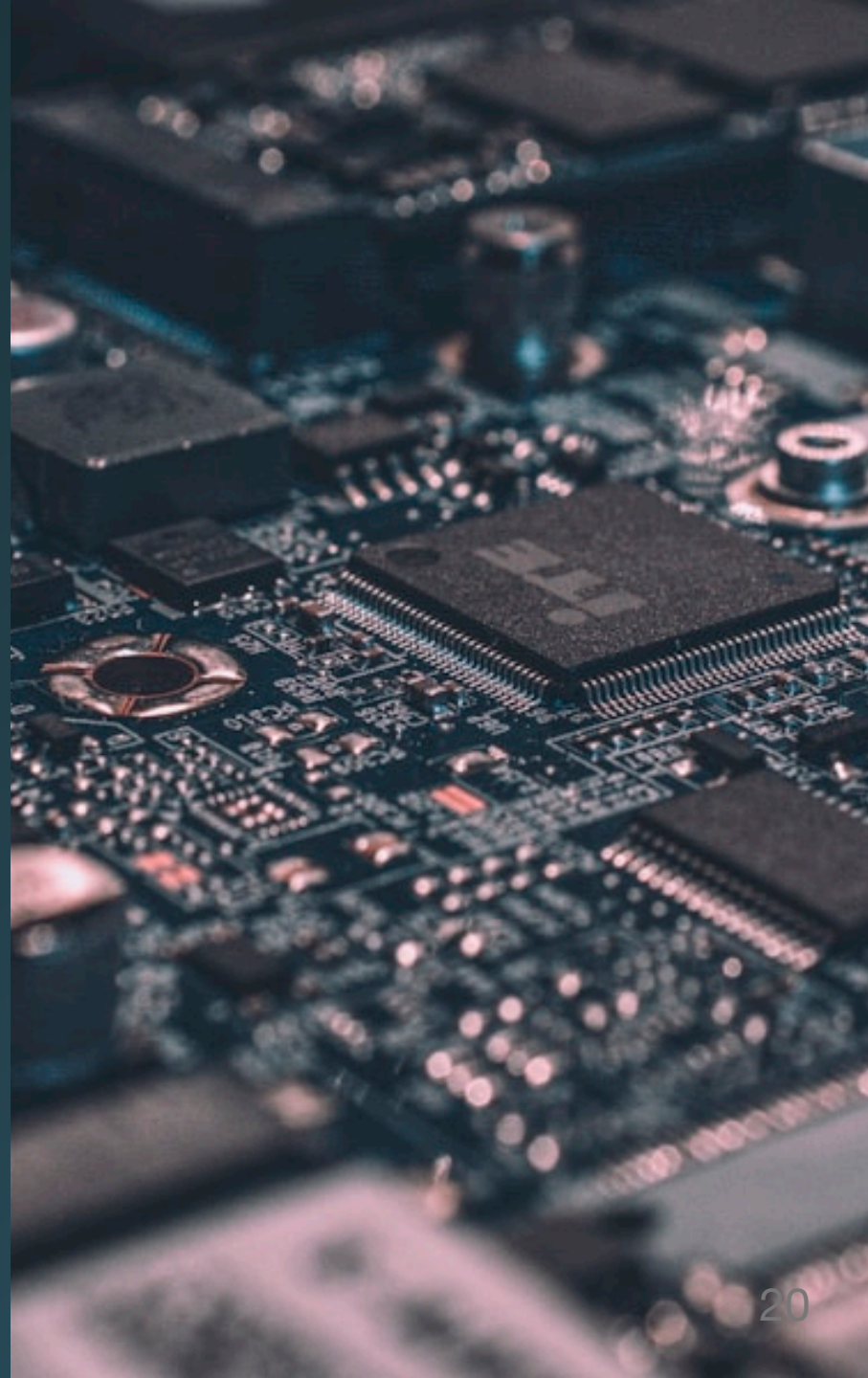
The Pipeline



One tool finds the pages. Another tool extracts the content. Your script glues them together.

Now the Upgrade: MCP

What if Claude Code could call Tavily and Firecrawl **directly**, without you writing Python?



Now the Upgrade: MCP

MCP servers (Model Context Protocol) extend Claude Code with new tools. Install one, and Claude Code can use it inside any prompt.

Python vs. MCP

🐍 Python (Part 02)

- ~35 lines of SDK calls
- Loops, file writes, sleeps
- Version-controlled
- Runs in GitHub Actions

💬 MCP (Part 03)

> "Find 5 URLs about Chipotle's recent earnings and save each as markdown in `knowledge/raw/`."

Same result. No code.

Today's Demo Domain: Chipotle IR

- **IR = Investor Relations** — shareholder-facing content
- Press releases, bios, earnings, filings
- Legally required to be accessible
- Great fit for a knowledge base

Accounts You Need

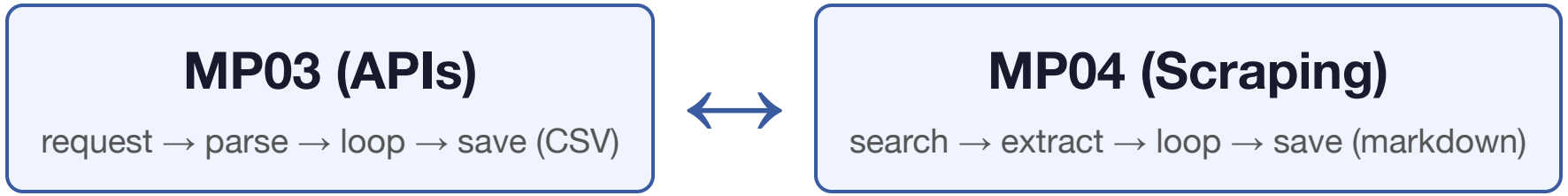
Service	URL	Free Tier
Firecrawl	firecrawl.dev	500 scrapes/month
Tavily	tavily.com	1,000 searches/month

Both: GitHub sign-in, no card. Under 2 minutes each.

Your Project Needs This

- 🎯 Milestone 02: **≥15 sources** from 3+ sites
- 🔗 Pattern works for any URL in a browser
- 💼 Your job posting says what content matters
- 📚 Feeds your knowledge base wiki

The Pattern Transfers



Different tools. Same shape. If you learned MP03, you already know MP04.

Next up: code it together.